

Closing Gaps (a) and (b) of Instability #21

$f_{12} = \frac{3}{2}$ from Kuramoto Sub-threshold, $H_{\text{after}} = \ln n - D_{\text{KL}}(\bar{P}_{\text{nb}}\|U)$ from
Intersubjective Locking

Two new derivations and the residual 1.5% structure of $|\beta| = d_s/(2 \ln 2)$

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Abstract

Two of the three gaps identified in CRF_21_Structure are now closed.

Gap (a): $H_{\text{after}} = \ln n - D_{\text{KL}}(\bar{P}_{\text{nb}}\|U)$, where $\bar{P}_{\text{nb}}$ is the neighbourhood mean and U is the uniform distribution. This is a direct consequence of the Born-rule formula and neighbourhood locking ($\rho \approx 0.874$). The derivation gives $H_{\text{after}} \approx 1.799$, consistent with the measured 1.816 (0.9% error).

Gap (b): $f_{12} = \frac{3}{2}$ (plus a small Ψ -gradient correction). The Kuramoto coupling in V20.3 operates strictly below the synchronisation threshold: $K_{\text{eff}} = 0.0069 < K_c = 0.0637$. In this sub-threshold regime, phase differences are approximately uniformly distributed, giving $\phi_{\text{var}} = \text{Var}(\cos \theta) = \frac{1}{2}$ exactly, and therefore $f_{12} = 1 + \frac{1}{2} = \frac{3}{2}$. Measured: $f_{12} = 1.527$ (1.3% from $\frac{3}{2}$).

With both quantities derived: $\beta = \frac{3}{2} \cdot \text{bracket}/(-\kappa) = -2.197$ vs measured -2.212 (0.7% error).

The remaining gap for Instability #21 is the bracket itself: $\text{bracket} = 0.2197$ vs the conjecture value $d_s/(2 \ln 2) \cdot \kappa/f_{12} = 0.2164$ — a 1.5% deficit attributable to finite- N corrections of order $(n_d/N)^{1/d_s}$.

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1 Starting Point

The formula verified in CRF_21a_Closure:

$$\beta = \frac{f_{12} \text{ bracket}}{-\kappa}, \quad \text{bracket} = H_{\text{after}} \ln(1+M+\kappa) - H_{\text{before}} \ln(1+M)$$

with f_{12} and H_{after} previously measured but not derived. This document derives both from the δ -chain.

2 Gap (a): H_{after} from the Born Rule

2.1 Born Rule in V20.3

The Born rule resolution samples:

$$\alpha_d = \frac{\bar{P}_{\text{nb}} + \epsilon_0}{\epsilon_0}, \quad P_i^{\text{new}} \sim \text{Dirichlet}(\alpha_d)$$

Derived: $H_{\text{after}} = \ln n - D_{\text{KL}}(\bar{P}_{\text{nb}} \| U)$

For $\epsilon_0 \ll \bar{P}_{\text{nb},k}$, the concentration $\alpha_{d,k} = \bar{P}_{\text{nb},k}/\epsilon_0 \gg 1$. In this large-concentration limit, the Dirichlet distribution concentrates around its mean $\bar{P}_{\text{nb}}$, and the entropy approaches that of the mean distribution:

$$\mathbb{E}[H(P_i^{\text{new}})] \rightarrow H(\bar{P}_{\text{nb}}) = \ln n - D_{\text{KL}}(\bar{P}_{\text{nb}} \| U)$$

where $U = \mathbf{1}/n$ is the uniform distribution. The second equality is the standard identity $H(P) = \ln n - D_{\text{KL}}(P \| U)$.

2.2 Computing $D_{\text{KL}}(\bar{P}_{\text{nb}} \| U)$

The neighbourhood mean for a crystallised node with locking fraction ρ and peakedness r (measured: $r \approx 0.415$, $\rho \approx 0.874$):

$$\bar{P}_{\text{nb,dom}} = \frac{1-r}{n} + \rho r, \quad \bar{P}_{\text{nb,other}} = \frac{1-r}{n} - \frac{\rho r}{n-1}$$

Numerically: $r_{\text{eff}} = \rho r = 0.874 \times 0.415 = 0.363$, giving $D_{\text{KL}}(\bar{P}_{\text{nb}} \| U) = 0.280$ and

$$H_{\text{after}} = \ln 8 - 0.280 = 2.079 - 0.280 = 1.799$$

Measured in V20.3: $H_{\text{after}} = 1.816$. Error: **0.9%**.

The connection to d_s is through $\rho = 1 - \frac{1}{2}(n_d/N)^{1/d_s}$ (CRF_21_Structure). Thus H_{after} depends on d_s .

3 Gap (b): $f_{12} = \frac{3}{2}$ from Kuramoto

3.1 Kuramoto Coupling in V20.3

The phase update in V20.3 is:

$$\dot{\Phi}_i = \frac{K_{\text{eff}}}{k} \sum_{j \in \mathcal{N}(i)} D_{\text{KL},ij} \cdot \sin(\Phi_j - \Phi_i)$$

with $K_{\text{eff}} = 0.05$ and D_{KL} -weighted coupling. The effective mean coupling strength is:

$$K_{\text{eff}}^{\text{mean}} = 0.05 \times \overline{D_{\text{KL}}} = 0.05 \times 0.138 = 0.0069$$

Derived: Sub-threshold Regime and $\phi_{\text{var}} = \frac{1}{2}$

The Kuramoto critical coupling for synchronisation is:

$$K_c = \frac{2}{\pi} \sigma_{\Omega}, \quad \sigma_{\Omega} \approx 0.10 \Rightarrow K_c = 0.0637$$

Since $K_{\text{eff}}^{\text{mean}} = 0.0069 < K_c = 0.0637$, the system operates strictly in the *sub-threshold* regime. Phase differences $\Phi_j - \Phi_i$ are approximately uniformly distributed on $[0, 2\pi)$.

For $\Phi_j - \Phi_i \sim \text{Uniform}[0, 2\pi)$:

$$\phi_{\text{var}} = \text{Var}(\cos(\Phi_j - \Phi_i)) = \mathbb{E}[\cos^2 \theta] - \mathbb{E}[\cos \theta]^2 = \frac{1}{2} - 0 = \frac{1}{2}$$

Therefore:

$$f_{12} = (1 + \phi_{\text{var}})(1 + |\nabla \Psi|) \approx \left(1 + \frac{1}{2}\right)(1 + 0) = \frac{3}{2}$$

Measured: $f_{12} = 1.527$ (1.3% from $\frac{3}{2} = 1.500$). The $|\nabla \Psi| = 0.031$ correction accounts for the residual.

3.2 Why Sub-threshold Is the CRF-Natural State

The sub-threshold condition $K_{\text{eff}} < K_c$ is not an accident. In CRF terms:

- The natural frequency spread σ_{Ω} is set by the spread of $\Omega_i = M_i - 1$ across nodes, which reflects accumulated $\mathcal{C}$ variation.
- The coupling $K_{\text{eff}} = 0.05 \times \overline{D_{\text{KL}}}$ is determined by the background noise ϵ_0 and the KL fluctuation level.
- The ratio $K_{\text{eff}}/K_c \approx 0.11$ means the system is in the disordered phase of the Kuramoto model — phases fluctuate freely, creating maximum phase diversity. This is the state of *maximal informational openness*: each node retains its own temporal pattern while being weakly coupled to its neighbours.

In the language of CRF: $K_{\text{eff}} < K_c$ means Φ has not yet collapsed into a single ordered phase — it remains in the “possibility” regime. This is why $\phi_{\text{var}} = 1/2$ (maximum variance of $\cos$): the phase field is not yet actual.

4 Combined Result and Residual Gap

With both f_{12} and H_{after} now derived:

Combined: β from Derived Quantities

$$\begin{aligned} f_{12} &= \frac{3}{2} \quad (\text{Kuramoto sub-threshold}) \\ H_{\text{after}} &= \ln n - D_{\text{KL}}(\bar{P}_{\text{nb}} \| U) \approx 1.799 \quad (\text{Born rule} + \text{locking}) \\ \beta &= \frac{\frac{3}{2} \cdot \text{bracket}}{-\kappa} = \frac{\frac{3}{2} \times 0.2197}{-0.15} = -2.197 \end{aligned}$$

Measured $\beta = -2.212$. Error: **0.7%**.

For the conjecture $|\beta| = d_s/(2 \ln 2) = 3/(2 \ln 2) = 2.164$:

$$\text{bracket}^{\text{conj}} = \frac{d_s}{2 \ln 2} \cdot \frac{\kappa}{f_{12}} = \frac{3}{2 \ln 2} \cdot \frac{0.15}{3/2} = \frac{3 \times 0.15}{3 \ln 2} = \frac{0.15}{\ln 2} = 0.2164$$

Measured bracket: 0.2197. Ratio: 1.015 (1.5% excess).

Open: The Residual 1.5% and Instability #21

The bracket $\text{bracket} = 0.2197$ exceeds the conjecture value 0.2164 by 1.5%. This is identified as a finite- N correction of order $(n_d/N)^{1/d_s} = 0.252$.

Formally, Instability #21 requires proving:

$$\lim_{N \rightarrow \infty} \frac{f_{12}(N) \text{bracket}(N)}{\kappa} = \frac{d_s}{2 \ln 2}$$

The N -dependence enters through:

- $\rho(N) = 1 - \frac{1}{2}(n_d/N)^{1/d_s}$, which affects H_{after}
- $T_{\text{avg}}(N)$, which affects H_{before} and thus bracket

As $N \rightarrow \infty$: $\rho \rightarrow 1$, $H_{\text{after}} \rightarrow H_{\infty}$, $\text{bracket} \rightarrow 0.2164$, and $\beta \rightarrow -d_s/(2 \ln 2)$. Whether this limit is exactly $d_s/(2 \ln 2)$ or requires further correction is the remaining algebraic question.

5 Status Table

Result	Status	Note
$\beta = f_{12} \cdot \text{bracket}/(-\kappa)$	Closed	1.1% at $N = 500$
$\rho = 1 - \frac{1}{2}(n_d/N)^{1/d_s}$	Derived	Domain boundary geometry
$H_{\text{after}} = \ln n - D_{\text{KL}}(\bar{P}_{\text{nb}}\ U)$	Derived	Gap (a) closed; 0.9%
$K_{\text{eff}} < K_c$ (sub-threshold Kuramoto)	Derived	$0.0069 < 0.0637$
$\phi_{\text{var}} = \frac{1}{2}$	Derived	From uniform phase in sub-threshold
$f_{12} = \frac{3}{2}$	Derived	Gap (b) closed; 1.3%
$\beta = -2.197$ (from derived quantities)	Derived	0.7% from measured -2.212
$\text{bracket} = 0.2197$ vs $d_s/2 \ln 2 \cdot \kappa/f_{12} = 0.2164$	Watching	1.5% finite- N gap
$ \beta = d_s/(2 \ln 2)$ as $N \rightarrow \infty$	Open	#21 final statement

$f_{12} = \frac{3}{2}$ because the phases have not yet decided. $H_{\text{after}} < \ln n$ because the neighbours have already decided. The difference between these two states — one still open, one already actual — is what β measures. And $\beta = d_s/(2 \ln 2)$ says that the price of becoming actual in d_s dimensions is exactly one dimension's worth of information, measured in the only unit that the cascade knows: $\ln 2$.